Power

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Compare how quickly energy is transferred.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 13.*
Difference 6 J may be rotational energy and thermal energy; other justified transfers acceptable.

02. Worked example: explain the model
Energy equal; shorter time means greater power.

03. Read and interpret the model
First; 60 W versus 20 W.

04. Apply the physics
P = 90/5 = 18 W. Over 6 s, E = 18 × 6 = 108 J.

05. Explain your reasoning
100 J per second; energy=power×specified duration.

06. Review: connect an earlier idea *Review: Unit 4, Day 4.*
W_{net}=-20 J; speed falls for fixed mass.

07. Exit check
Same work; faster student twice average power.

08. STRETCH: test the model
Examples 100 J in 2 s and 250 J in 5 s.

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01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 13.*
mgh≈1.47 J; K=0.3375 J; fraction≈0.2296 (22.96%).

02. Worked example: explain the model
Energy equal; shorter time means greater power.

03. Read and interpret the model
First; 60 W versus 20 W.

04. Apply the physics
2352 J; 294 W.

05. Explain your reasoning
100 J per second; energy=power×specified duration.

06. Review: connect an earlier idea *Review: Unit 4, Day 4.*
$W_{net}=\frac{1}{2}m(v_f^2-v_i^2)$=-56 J, negative as kinetic energy falls.

07. Exit check
600 W; 3000 J.

08. STRETCH: test the model
Useful 150 W; input 200 W; input 800 J, useful 600 J, other 200 J.